Ch. 5 Currency Derivatives

-> Forwards
-> Futures
-> Options

Forwards / Futures: very similar. Forwards are customizable, Futures are standardized

Derivatives are financial products whose value is DERIVED from the value of another asset called the underlying

-> underlying: currency. EUR, GBP, JPY
-> derivative: forward contract specifying price of currency bought or sold in the future

Example:

Today you enter into forward contract to buy €100,000 at $1.10/€ in 3 months.

$S_0 = \$1.11/€$

Today: you negotiate terms and sign contracts. ☺ to minimal security deposit (deposit amount dependent on company history)

3 months: you have to buy €100,000 at $1.10/€

$S_3 = \$1.12/€$ or $S_3 = \$1.08/€$

Forward Rate: $F = \$1.10/€$

rate at which you will purchase in the future

Spot Rate: exchange rate at a given point in time.

⇒ Companies use derivatives contract

→ hedge exchange rate risk

→ speculation - benefiting from projected

changes in exchange rates.

→ if you think currency will go up, you can lock into price today to exchange in the future (buy at lower rate)

→ if you think currency value will go down, you can enter to sell forward contract in the future at a rate

specified today.

Futures and forwards trade at premium or discount

$$F = S_0 \cdot (1+p)$$

$$p = \frac{F}{S_0} - 1$$

$S_0 = \$1.681$ / GBP

	F	Maturity	P
1 month fwd	\$1.680	1 month	−0.71%
3 month fwd	\$1.677	3 months	
6 month fwd	\$1.672	6 months	

$$\frac{F}{S} - 1 = p \quad \text{per period}$$

Annualized

$$\frac{1.680}{1.681} - 1 \cdot \left(\frac{360}{30}\right) =$$

$$= -0.0595 \cdot 12 = -0.71\%$$

3 month forward p

$$p = \left[\frac{1.677}{1.681} - 1\right] \cdot \left(\frac{360}{90}\right)$$

$$= (0.9976 - 1) \cdot 4$$

$$= -0.95\%$$

Exhibit 5.2 Currency Futures Contracts Traded on the Chicago Mercantile Exchange

CURRENCY	UNITS PER CONTRACT
Australian dollar	100,000
Brazilian real	100,000
British pound	62,500
Canadian dollar	100,000
Chinese yuan	1,000,000
Czech koruna	4,000,000
Euro	125,000
Hungarian forint	30,000,000
Indian rupee	5,000,000
Israeli shekel	1,000,000
Japanese yen	12,500,000
Korean won	125,000,000
Mexican peso	500,000
New Zealand dollar	100,000
Norwegian krone	2,000,000
Polish zloty	500,000
Russian ruble	2,500,000
South African rand	500,000
Swedish krona	2,000,000
Swiss franc	125,000

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You want to buy 1 futures contract for Australian Dollar (AUS). Futures price for delivery in March is $0.53/AUS

Today: January 10: buy 1 futures contract

1 contract: 100,000 AUS

Value of contract = $0.53/AUS · 100000 AUS

= $53,000.00 USD

No money is exchanged today, other than small margin (initial margin)

January 12: F: $0.45/AUS

Value of contract $45,000

→ no money exchanged but your unrealized profit: $45,000 − 53

Prices change daily, if your losses = −$8000 exceed maintenance margin requirement, you may needs to add more funds to account

In February you decide, you want to close this position

→ you sell 1 futures contract with the same delivery → offset contract

→ $F_{March} = \$0.48/AUS$

Sell in March 100000 AS · $0.48/AS

= $48,000

→ Receive $48,000 in March

No cash transaction today

→ Third Thursday in March

You receive	$48,000
you pay	−$53,000
	−$5,000

loss

Exhibit 5.3 Comparison of the Forward and Futures Market

	FORWARD	FUTURES
Size of contract	Tailored to individual needs.	Standardized.
Delivery date	Tailored to individual needs.	Standardized.
Participants	Banks, brokers, and multinational companies. Public speculation not encouraged.	Banks, brokers, and multinational companies. Qualified public speculation encouraged.
Security deposit	None as such, but compensating bank balances or lines of credit required.	Small security deposit required. Margin
Clearing operation	Handling contingent on individual banks and brokers. No separate clearinghouse function.	Handled by exchange clearinghouse. Daily settlements to the market price.
Marketplace	Telecommunications network.	Globex computerized trading platform with worldwide communications.
Regulation	Self-regulating.	Commodity Futures Trading Commission; National Futures Association.
Liquidation	Most settled by actual delivery; some by offset, but at a cost.	Most by offset; very few by delivery.
Transaction costs	Set by the spread between bank's buy and sell prices.	Negotiated brokerage fees.

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